

# Technical Description

Gas Genset

**Control system MIP / MMC4000 from V3.5 onwards**

**MTU 8V4000 GS**

**MTU 12V4000 GS**

**MTU 16V4000 GS**

**MTU 20V4000 GS**

**MS61015/00E**

| SLU Name      | Sales Name              |  |
|---------------|-------------------------|--|
| GG08V4000A1   | MTU 8V4000 GS           |  |
| GG08V4000A2   | MTU 8V4000 GS           |  |
| GG12V4000A1   | MTU 12V4000 GS          |  |
| GG12V4000A2   | MTU 12V4000 GS          |  |
| GG16V4000A1   | MTU 16V4000 GS          |  |
| GG16V4000A2   | MTU 16V4000 GS          |  |
| GG20V4000A1   | MTU 20V4000 GS          |  |
| GG20V4000A2   | MTU 20V4000 GS          |  |
| Not valid for | xV4000L62 (natural gas) |  |
| Not valid for | xV4000L63 (natural gas) |  |

*Table 1: Applicability*

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# 1 Control

## 1.1 General information



All options listed can be acquired at an additional charge.

The series 4000 control system consists of the control cabinets MMC (MTU Module Control) and MIP (MTU Interface Panel). The MMC control cabinet is attached to the system separately on the floor. The MIP is mounted permanently on the genset frame and thus forms a functional unit.

### **The MMC essentially serves**

- for operation and visualization
- for activation of external auxiliary drives

### **The MIP essentially serves**

- for communication with engine governor ECU and engine monitoring unit EMU
- for synchronization and connection of the generator to mains
- for activation of auxiliary drives on the genset
- for generator and mains protection functions



The MIP can also be operated without MMC, however in this case the MMC functions must be implemented in a customer-side control system.

## 1.2 Operating modes

### **Mains parallel operation**

Upon getting a start command, the genset is started, and when all the operating conditions are reached, it is automatically synchronized to mains. Following this the specified electric power output is started up via a ramp.

### **Mains failure / standby operation (option)**

Since standard equipment of the MTU genset includes a synchronous generator, mains failure operation can be realized. When the mains fails, the plant takes over the isolated load (the maximum power circuit breakers of the respective engines are to be observed). Once the power has been restored and the settling period has lapsed, the system is automatically re-synchronized with the mains power. Determine the appropriate operating mode according to project requirements; the standard equipment of the control system will be accordingly extended with options.

### **Control mode (only in mains parallel operation)**

Adjustable value:

The power setpoint is specified variably from outside between a minimum / and a maximum value.

For optional operating modes see chapter "Peripheral auxiliary drives and functions".

## 1.3 MTU Interface Panel (MIP)

### 1.3.1 General information

#### **Essentially, the MIP contains the following components**

- Operating elements (EMERGENCY STOP button, main switch)
- PLC – central unit (Computer Programmable Control Unit with various interfaces and input/output modules)
- EMM (generator and mains protection, synchronizer) BDEW conformity
- Communication with engine governor ECU and engine monitoring unit EMU via hardware signals and CAN bus
- Interfaces for connecting to external systems (floating contacts)
- Activation of the auxiliary drives installed on the genset in the factory

The control program runs independently in the PLC central unit. This makes it possible to continue operating with the previously set parameters if the IPC fails.

### 1.3.2 Functions of the programmable logic controller (PLC)

The control program runs in the PLC. All signals (analog, digital) are read in and processed. Software updates are possible on site, via remote diagnosis or by exchanging the flash card.

Integrated functions:

- Recording of the measured data (analog and digital) of the genset
- Start-/stop procedures
- Control of the internal auxiliary drives of the genset
- Controls (power controller)
- Evaluation of all fault and status messages
- Monitoring of analog values
- Saving the parameters, measurement and operating data

### 1.3.3 Hardware control elements

- EMERGENCY STOP button
- Main switch, auxiliary drives



### 1.3.4 Synchronization and monitoring functions

#### Monitoring functions

The Energy Measurement Module is integrated in the PLC system (supply voltage, communication with the PLC, parameterization). However, the Energy Measurement Module has a separate controller to carry out the monitoring and synchronizing functions. All generator and mains signals (currents, voltages, etc.) can be read into the Energy Measurement Module. The recorded data are used to implement protective functions and the synchronization.

#### Generator protection

- Overfrequency (ANSI: 81, IEC: f>)
- Underfrequency (ANSI: 81, IEC: f<)
- Overvoltage (ANSI: 59, IEC: U>)
- Undervoltage (ANSI: 27, IEC: U<)
- Normally dependent overcurrent (ANSI: 51, IEC: I>t>)
- Overcurrent (ANSI: 50, IEC: I>>)
- Overload (ANSI: 32F, IEC: Pf>>)
- Reverse power (ANSI: 32R, IEC: Pr>)
- Unbalanced load (ANSI: 46, IEC: I2>)

#### Synchronization function

- Synchronization (ANSI: 25, IEC: —)

### 1.3.5 Interfaces

The MIP is equipped with the following standard interfaces for external connection:

- Hardware interface (e.g. floating contacts, measurement voltages) for connection to the periphery.
- Bus interface for connection to the central common control system (option)

#### Hardware interface

Input and output signals are available at the hardware interface (terminal strip).

Input signals:

- Mains voltage (synchronization / mains protection)
- Bus bar voltage (synchronization / mains protection)
- Generator circuit breaker feedback (on, off, fault)
- Mains circuit breaker feedback (on, off, fault)
- Mains protection from outside / MMC
- Power failure test from outside / MMC
- Mains disconnect through mains circuit breaker from outside / MMC
- Manual EMERGENCY STOP feedback circuit from outside / MMC
- Machine EMERGENCY STOP feedback circuit from outside / MMC
- Reset safety chains from outside / MMC
- Clearance for gas pressure min. delayed after start request of gas compressor (only for biogas) from outside / MMC
- Differential protection, ground fault protection from outside / MMC
- Generator voltage at medium voltage
- Measurement voltage between step-up transformer and generator circuit breaker (depending on the electrical connection), optional
- Measurement current between step-up transformer and generator circuit breaker (depending on the electrical connection), optional

Output signals:

- Activation generator circuit breaker (on, off)
- Actuation of mains power switch (on, off)
- Start request for auxiliary drives, general
- Open back-up switch
- Manual EMERGENCY STOP to outside / MMC
- Machine EMERGENCY STOP to outside / MMC

#### Bus interface

- See chapter "Peripheral auxiliary drives and functions"

#### Remote diagnosis interface

- See chapter "Peripheral auxiliary drives and functions"